$Cu\,(s)\left|Cu^{2+}(0.25\ aq,\ M)\right\|Fe^{3+}(0.005\ aq\ M)\left|Fe^{2+}(0.1\ aq\ M)\right|Pt\,(s)$

Given : $\left(E^{\circ}\right)_{Fe^{3+}|Fe^{2+}} = 0.77V$ and $\left(E^{\circ}\right)_{Cu^{2+}|Cu} = 0.34\ V$

Half reactions are

$Cu\,(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$ ..... (1)

$2\,Fe^{3+}(aq) + 2e^{-} \rightarrow 2Fe^{2+}(aq)$ ..... (2)

the overall reaction is

$Cu\,(s) + 2\,Fe^{3+}(aq) \rightarrow Cu^{2+}(aq) + 2\,Fe^{2+}\,(aq)$, and $n = 2$

Apply Nernst equation at .25°C .

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{2}\log\frac{[Cu^{2+}][Fe^{2+}]^2}{[Fe^{3+}]^2} \qquad [\because\ [Cu\,(s)] = 1]$$

$$E^{\circ}_{cell} = \left(E^{\circ}_{ox}\right)_{Cu|Cu^{2+}} + \left(E^{\circ}_{red}\right)_{Fe^{3+}|Fe^{2+}}$$

Given standard reduction potential of $Cu^{2+}|Cu$ is 0.34V

$\therefore\ \left(E^{\circ}_{ox}\right)_{Cu|Cu^{2+}} = -0.34V$

$\left(E^{\circ}_{red}\right)_{Fe^{3+}|Fe^{2+}} = 0.77V$

$\therefore\ E^{\circ}_{cell} = -0.34 + 0.77$

$E^{\circ}_{cell} = 0.43V$

$$\therefore\ E_{cell} = 0.43 - \frac{0.0591}{2}\times\log\frac{(0.25)(0.1)^2}{(0.005)^2}$$

$$= 0.43 - \frac{0.0591}{2}\times 2$$

$$= 0.43 - 0.0591$$

$$= 0.3709V.$$

$$= \log\frac{(0.25)(0.1)^2}{(0.005)^2}$$

$$= \log\frac{25\times10^{-2}\ \times\ 1\times10^{-2}}{25\times10^{-6}}$$

$$= \log 10^2$$

$$= 2\log_{10}10$$

$$= 2.$$

Evaluate yourself

The electrochemical cell reaction of the Daniel cell is

$Zn\,(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu\,(s)$

What is the change in the cell voltage on increasing the ion concentration in the anode compartment by a factor 10?

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